

Research Proposal - Alberto Almuna Morales

Background

There is a consensus that, to achieve an effective outcome in therapeutic ultrasound, a patient-specific planning phase must be carried out beforehand. The patient's anatomical information can be obtained from biomedical scans, which allow for personalized modelling of the patient for the treatment. Given this, it is necessary to have fast, accurate, and efficient solvers that enable the correct simulation of the treatment on the patient based on their anatomical information, and that can be used to verify the feasibility, safety, and precision of the treatment.

Histotripsy is a non-invasive medical treatment that uses high-intensity focused ultrasound waves to treat liver tumours. Currently, the feasibility of applying the treatment to a patient depends on the location of the tumour. If the path of the ultrasound waves is clear and without interference, the patient can be treated. Conversely, if the tumour is located near the ribs, its treatment is not feasible, and the patient is excluded from the treatment. The reason for this is due to the difficulties in predicting what will happen to the ultrasound waves when interacting with bone.

The simulation of high-intensity focused ultrasound through biological tissues is characterized by the correct modelling of the nonlinear behaviour of the waves and the consideration of the heterogeneous properties of the materials. Although nonlinear acoustic models exist, solving nonlinear wave propagation in large three-dimensional heterogeneous domains remains computationally challenging.

Many therapeutic ultrasound problems can be simulated in the frequency domain. The Boundary Element Method (BEM) implemented in OptimUS for simulating focused ultrasound with piecewise homogeneous domains has proven to be a promising alternative due to its reduction in computation time compared to other software and its low memory requirements. The team has already extended the methodology to incorporate heterogeneous materials through volume and surface integral equation (VSIE) methods, and weakly nonlinear problems for low-contrast media scenarios. Given the above, there is potential to further extend these methodologies for emerging focused ultrasound applications that exhibit strongly nonlinear behaviour, such as histotripsy.

Research question

How do rib bone and heterogeneous tissues influence patient-specific planning for ultrasound ablation therapy of liver tumours?

Proposal

Integral equation methods can be used to model nonlinear ultrasound propagation through high-contrast interfaces (like bone) and heterogeneous tissues in the frequency domain. The methodology will help us understand the conditions under which nonlinear, heterogeneous modelling is essential for accurate treatment planning in ultrasound therapies such as histotripsy.

Scope

- Therapeutic ultrasound: Literature review on current and emerging ultrasound therapies to understand the stage of adoption and identify current gaps/challenges.
- Physical model: Description of the physics of linear and nonlinear ultrasound propagation in biological media, including soft tissue and bone, in the form of time dependant differential equations.
- Harmonic model: Expression of the time-dependent partial differential equation as a finite series of harmonic frequencies.
- Numerical method: Integral equation approach to solve nonlinear wave propagation in heterogeneous tissue, high-contrast interfaces and for multiple harmonics.
- High-performance computing: Study of different implementation architectures (CPU and GPU), algorithms (fast multipole methods and hierarchical matrix compression) and programming languages (Python with Numba, OpenCL, Rust and C) to design fast algorithms to solve the VSIE.
- Software engineering: Incorporation in the Optimus software library.

Timeline

1st Package:

Focused ultrasound for thermal ablation with homogeneous materials using frequencies, amplitudes and parameters similar to the work “Accelerating frequency-domain numerical methods for weakly nonlinear focused ultrasound using nested meshes” (Samuel P. Groth, et al.).

Activities:

- Equation derivation [M4-M7]: Derive the complete physical equation for the nonlinear ultrasonic waves in heterogeneous domains. The equation must consider all relevant considerations for clinically relevant treatment-planning scenarios (e.g.: thermo-viscous attenuation, molecular relaxation, power-law attenuation and nonlinearities). Identify the correct interface conditions for high-contrast interfaces, like the boundary between bone and water.

- Multi-harmonic formulation [M7]: Write the multi-harmonic formulation of the time-dependent PDE using a process similar to Sam's work. Model power-law attenuation with a fractional derivative.
- Numerical method [M8-M9]: Apply VSIE as a numerical method for models with homogeneous rib bone, and homogeneous water as exterior domain (piecewise constant material parameters). Use a bowl transducer and a phased array transducer as the source with 1 MHz.
- High-performance computing [M10-M11]: Design an efficient VSIE implementation with nested meshes. Similar to Sam's work but now incorporating bone (high-contrast media). Possibly include hierarchical matrices.

Deliverables:

- Paper: Simulating weakly-nonlinear focused ultrasound behind the ribcage [M4-M13].
- Report for the PhD upgrade [M4-M13].

2nd Package:

Inclusion of heterogeneous media for ultrasound ablation behind the ribcage.

Activities:

- Equation derivation: Determine whether it is necessary to include higher-order terms for the gradients of ambient state variables in the case of heterogeneous materials. Study the differences between media with low and high local variations in their properties, as well as their impact on the approximation scheme of the equation.
- Multi-harmonic formulation: Update the multi-harmonic formulation in line with the new terms in the equation.
- Numerical method: Extend the previous VSIE numerical method to piecewise heterogeneous domains; specifically, consider a heterogeneous rib bone surrounded by water with constant properties and the same transducers as before.
- High-performance computing: Study the computational feasibility of considering heterogeneous material properties in the efficient VSIE implementation and update it accordingly.
- Impact quantification: Quantify the importance of heterogeneous modelling for treatment planning. Compare homogeneous and heterogeneous simulations with low and high local variations in their material properties, and study their impact on the formation of the ultrasound focus.

Deliverables:

- Paper about heterogeneous focused ultrasound simulations in the ribcage, and comparisons between simulations involving different types of heterogeneities and their impact on the formation of the ultrasound focus.

3rd Package:

Strong nonlinearity and higher harmonics.

Activities:

- Equation derivation: Investigate whether the current equation is valid for the strongly nonlinear case. Possibly include higher-order terms in the approximation scheme.
- Multi-harmonic formulation: Study how many more harmonics should be included in the strongly nonlinear case and add them to the formulation.
- Numerical method: Adapt the VSIE numerical method and possibly implement a hybrid approach, combining full-wave solvers with high-frequency techniques. Consider again a rib bone surrounded by water model, and the same transducers as before but with shorter pulses and bigger amplitudes.
- High-performance computing: Investigate the feasibility of solving highly non-linear problems in the frequency domain and incorporate high-performance computing techniques to make the model faster and more efficient. If other high-frequency techniques are included, investigate how to combine them with the full-wave solver.
- Impact quantification: Quantify the importance of nonlinear modelling for treatment planning of ultrasound ablation therapies. Compare linear, weakly and strongly nonlinear simulations with low and high number of harmonics. Identify when more complex approaches are needed. The aim is to investigate what amplitude values are required in the transducers to achieve the operating thresholds for ultrasound ablation therapies (e.g. cavitation histotripsy) with the different models.

Deliverables:

- Paper about impact quantification of the importance of nonlinear and heterogeneous modelling, and the number of harmonics to consider for treatment planning in ultrasound ablation therapy of liver tumours.
- Paper about high-performance computing for efficient VSIE in nonlinear focused ultrasound.

4th Package: (last year)

Finish up PhD work and explore extensions of our methodology.

Activities:

- PhD Thesis [M37-M42]: Write my PhD thesis.
- OptimUS [M43-M45]: Consolidating the integration into OptimUS.
- Extensions [M46-M48]: Explore extensions towards related ultrasound therapies, like ablation, neuromodulation and blood-brain barrier opening. Additionally, investigate the influence of air-filled parts of the body, such as the stomach and lungs. Study the impact quantification of these treatments.

Deliverables:

- PhD thesis [M42].
- New numerical methodology fully implemented in OptimUS [M45].
- Paper about the extensions of our work, validation in other focused ultrasound therapies and comparisons with other software [M48].
- * Review paper on nonlinear ultrasound ablation in biomedical tissue [In parallel with the progress of the PhD].

Novelty

Development of a numerical method for high-intensity nonlinear focused ultrasound in high-contrast and heterogeneous media using volume and surface integral equations methods. Additionally, studying the impact of different modeling approaches will help identify the capabilities and limitations of various ultrasound ablation treatments, quantifying the precision gained in patient-specific simulation and treatment planning through the application of more complex methodologies.